

Test Case ID	Test Description
File I/O & Exception Handling	
FILE-01	CSV file loads successfully
FILE-02	CSV file not found
FILE-03	CSV permission error
FILE-04	JSON file written on quit
FILE-05	JSON reflects updated data
FILE-06	JSON write error handled gracefully
Menu & Control Flow	
MENU-01	Quit immediately
MENU-02	Invalid menu option
MENU-03	Lowercase menu option accepted
MENU-04	Loop continues after action
List Option (L)	
LIST-01	List all servers — default (no key)
LIST-02	List by valid key: hostname

LIST-03	List by valid key: domain
LIST-04	List by valid key: ipv4
LIST-05	List by valid key: port
LIST-06	List by valid key: location
LIST-07	List by invalid key
Search Option (S)	
SRCH-01	Search — single match
SRCH-02	Search — multiple matches
SRCH-03	Search — match on IP value
SRCH-04	Search — no match
SRCH-05	Search — case insensitive input
Add Option (A)	
ADD-01	Add server — all attributes provided
ADD-02	Add server — verify persists to JSON
ADD-03	Add server — blank/partial attributes
ADD-04	Add multiple servers sequentially
Delete Option (D)	

DEL-01	Delete server — valid FQDN exact match
DEL-02	Delete server — valid FQDN partial match
DEL-03	Delete server — invalid FQDN
DEL-04	Delete server — verify persists to JSON
Iteration	
ITER-01	Quit immediately (single iteration)
ITER-02	Quit after one operation
ITER-03	Quit after multiple operations
Data Integrity	
DATA-01	All CSV rows read as list of dicts
DATA-02	JSON output contains updated server data

inventory_servers.py — Test Pla

Input / Steps
Run program with valid hosts.csv present
Rename hosts.csv, then run program
Remove read permission on hosts.csv (chmod 000), run program
Run program, select Q immediately
Add a server, delete a server, then quit
Make hosts.json a read-only directory, then quit
Enter menu option: Q
Enter menu option: Z
Enter menu option: l (lowercase)
Perform any action (e.g. L), then Q
Select L; press Enter (blank criteria)
Select L; enter criteria: hostname

Select L; enter criteria: domain

Select L; enter criteria: ipv4

Select L; enter criteria: port

Select L; enter criteria: location

Select L; enter criteria: invalidkey

Select S; enter criteria: imap

Select S; enter criteria: happy.lab

Select S; enter criteria: 172.16.30.101

Select S; enter criteria: zzznomatch

Select S; enter criteria: HAPPY (uppercase)

Select A; enter: www / peachy.lab / 172.16.31.443 / 443 / wt206

Add server as above, then quit

Select A; leave one or more fields blank

Select A twice, adding two different servers

Select D; enter FQDN: www.happy.lab

Select D; enter FQDN: dns (partial)

Select D; enter FQDN: notaserver.fake.lab

Delete a server, then quit

Enter Q at first prompt

Perform L, then Q

Perform L, S, A, D in sequence, then Q

Run program; select L with blank criteria

Add one server, delete one server, quit

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Expected Output	Pass / Fail
Server data loaded; menu displayed	Pass
"File not found. Exiting program." printed; program exits	Pass
"Permission error. Exiting program" printed; program exits	Pass
hosts.json created with all original server records in JSON array format	Pass
hosts.json contains added server; deleted server absent	Pass
"Error writing JSON file:" message printed; program does not crash	Pass
"Exiting server management program." printed; hosts.json created	Pass
"Invalid menu option." printed; menu re-displayed	Pass
Program treats input as L; listing workflow begins	Pass
Menu re-displayed after action; program exits on Q	Pass
All server dictionaries printed, one per line	Pass
Each line shows "hostname : <value>" for every server	Pass

Each line shows "domain : <value>" for every server	Pass
Each line shows "ipv4 : <value>" for every server	Pass
Each line shows "port : <value>" for every server	Pass
Each line shows "location : <value>" for every server	Pass
Each line shows "invalidkey : None" for every server (dict.get returns None)	Pass
One server record printed (imap.peachy.lab)	Pass
All servers in happy.lab domain printed	Pass
Server www.happy.lab printed	Pass
"No servers found." printed	Pass
Input lowercased by get_search_string(); matches happy.lab servers	Pass
"Added {'hostname': 'www', ...}" printed; new record in server list	Pass
New server record present in hosts.json	Pass
Server added with empty string values; no crash (validation out of scope)	Pass
Both servers appear in subsequent listing and in hosts.json	Pass

Server removed; subsequent listing does not include www.happy.lab	Pass
First server whose hostname+domain contains 'dns' is removed	Pass
"Server record does not exist." printed; list unchanged	Pass
Deleted server absent from hosts.json	Pass
Program exits after writing JSON; no menu loop entered	Pass
Menu shown once after L; program exits cleanly on Q	Pass
Each action completes correctly; program exits; JSON reflects all changes	Pass
Number of rows printed equals number of data rows in hosts.csv	Pass
hosts.json row count = original + 1 - 1 = original count	Pass